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Author(s): ANDREAS FUSTER, PAUL GOLDSMITH-PINKHAM, TARUN RAMADORAI and ANSGAR WALTHER

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Predictably Unequal? The Effects of Machine Learning on Credit Markets

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ABSTRACT

Innovations in statistical technology in functions including credit-screening have raised concerns about distributional impacts across categories such as race. Theoretically, distributional effects of better statistical technology can come from greater flexibility to uncover structural relationships or from triangulation of otherwise excluded characteristics. Using data on U.S. mortgages, we predict default using traditional and machine learning models. We find that Black and Hispanic borrowers are disproportionately less likely to gain from the introduction of machine learning. In a simple equilibrium credit market model, machine learning increases disparity in rates between and within groups, with these changes attributable primarily to greater flexibility.

IN RECENT YEARS, NEW PREDICTIVE statistical methods and machine learning techniques have been rapidly adopted by businesses seeking profitability gains in a broad range of industries.¹ The pace at which these technologies have been adopted has prompted concerns that the risks associated with their

Andreas Fuster is at EPFL, Swiss Finance Institute, and CEPR. Paul Goldsmith-Pinkham is at Yale School of Management. Tarun Ramadorai is at Imperial College London and CEPR. Ansgar Walther is at Imperial College London. We thank Amit Seru (the Editor) and three anonymous referees for thoughtful comments. We also thank to Tobias Berg, Philippe Bracke, Jediphi Cabal, John Campbell, Francesco D'Acutto, Andrew Ellul, Kris Gerardi, Andra Ghent, Johan Hombert, Ralph Koijen, Andres Liberman, Gonzalo Maturana, Adair Morse, Karthik Muralidharan, Daniel Paravisini, Jonathan Roth, Jann Spiess, Jeremy Stein, Daniel Streitz, Johannes Stroebel, Boris Vallee, Stijn Van Nieuwerburgh, and participants at numerous conferences and seminars for helpful discussions and comments. We thank Kevin Lai, Lu Liu, and Qing Yao for research assistance. Fuster and Goldsmith-Pinkham were employed at the Federal Reserve Bank of New York, while much of this work was completed. The views expressed are those of the authors and do not necessarily reflect those of the Federal Reserve Bank of New York or the Federal Reserve System. In compliance with *The Journal of Finance* disclosure policy, we have no conflicts of interest to disclose.

Correspondence: Tarun Ramadorai, Imperial College, Exhibition Road, South Kensington, London SW7 2AZ and CEPR; e-mail: t.ramadorai@imperial.ac.uk.

¹ See, for example, Agrawal, Gans, and Goldfarb (2018). Academic economists also increasingly rely on such techniques (e.g., Belloni, Chernozhukov, and Hansen (2014), Varian (2014), Kleinberg et al. (2018a), Mullainathan and Spiess (2017), Chernozhukov et al. (2017), Athey and Imbens (2017)).

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use have not been carefully evaluated, including the possibility that any gains arising from better statistical modeling may not be evenly distributed.² In this paper, we study the distributional consequences of the adoption of machine learning techniques in the domain of household credit markets. We do so by developing basic theoretical frameworks to analyze these issues, conducting empirical analysis on a large administrative data set of loans in the U.S. mortgage market, and undertaking an initial assessment of potential economic magnitudes using a simple equilibrium model.

The essential idea underlying our paper is that a more sophisticated statistical technology (in the sense of reducing predictive mean-squared error [MSE]) produces predictions with greater variance than a more primitive technology. When applied to the context we study, our insight is that improvements in predictive technology act as mean-preserving spreads for predicted outcomes—in our application, predicted default propensities on loans.³ This means that some borrowers will always be considered less risky by the new technology (“winners”), while other borrowers will be deemed riskier (“losers”), relative to their position under the preexisting technology. The key question therefore is how these winners and losers are distributed across important categories such as race, income, or gender.

We attempt to provide clearer guidance to identify the specific groups most likely to win or lose from the change in technology. To do so, we first consider the decision of a lender who uses a single exogenous variable (e.g., a borrower characteristic such as income) to predict default. We find that who wins or loses depends on both the functional form of the new technology and the differences in the distribution of the characteristics across groups. Perhaps, the simplest way to understand this point is to consider an economy endowed with a primitive prediction technology that simply uses the mean level of a single characteristic to predict default. In this case, the predicted default rate will be the same for all borrowers, regardless of their particular value of the characteristic. If a more sophisticated linear technology that identifies that default rates are linearly decreasing in the characteristic becomes available to this economy, groups with below-average values of the characteristic will clearly be penalized following the adoption of the new technology, while those with higher values will benefit from the change. Similarly, a convex quadratic function of the underlying characteristic will penalize groups with higher variance of the characteristic, and so forth.

To extend this simple theoretical intuition, we describe two important mechanisms through which such unequal effects could arise. Default outcomes can generically depend on both “permissible” observable variables such as income or credit scores, and “restricted” variables such as race or gender. We focus on the case in which lenders are prohibited from using the latter set of variables

² See, for example, O’Neil (2016), Hardt, Price, and Srebro (2016), Kleinberg, Mullainathan, and Raghavan (2016), and Kleinberg et al. (2018b).

³ Academic work that applies machine learning to credit risk modeling includes Khandani, Kim, and Lo (2010) and Sirignano, Sadhwani, and Giesecke (2021).

to predict default but can freely apply their available technology to the permissible variables. In this setting, the first *flexibility* mechanism that we consider is that the additional flexibility available to the more sophisticated technology allows it to more easily recover the structural mapping between permissible variables and default outcomes. Another possible mechanism, *triangulation*, is that the structural relationship between permissible variables and default is perfectly estimated by the primitive technology but the more sophisticated technology can triangulate the effect of the unobserved restricted variables on the outcome by more effectively and accurately combining the observed permissible variables. In the latter case, some groups may be penalized or rewarded based on realizations of the permissible variables, as the more sophisticated technology “de-anonymizes” group identities in the data using the permissible variables.⁴

Our theoretical work is helpful to build intuition, but credit default forecasting generally uses a large number of variables and machine learning involves highly nonlinear functions. This means that it is not easy to identify general propositions about the joint distributions of characteristics across groups or the functional form predicting default. Indeed, the impact of new technology could be negative or positive for any given group of households. For instance, there are numerous real-world examples of new entrants with more sophisticated technology more efficiently screening and providing credit to members of groups that were eschewed by those using more primitive technologies.⁵ Armed with the intuition from our simple models, we therefore go to the data to understand the potential effects of machine learning on an important credit market, namely, the U.S. mortgage market.

In our empirical work, we rely on a large administrative data set of close to 10 million U.S. mortgages originated between 2009 and 2013 in which we observe borrowers’ race, ethnicity, and gender as well as mortgage characteristics and default outcomes.⁶ Using these data, we estimate a set of increasingly sophisticated statistical models to predict default. The models begin with a logistic regression of default outcomes on borrower and loan characteristics and culminate in a Random Forest (RF) machine learning model (Ho (1998), Breiman (2001)).⁷

We confirm that the machine learning technology delivers statistically significantly higher out-of-sample predictive accuracy for default than the simpler

⁴ While the concept of triangulation has been well investigated by prior work in the area (e.g., Ross and Yinger (2002), Pope and Sydnor (2011)), we add to this line of research by investigating how the incidence of triangulation is affected by the introduction of more sophisticated prediction technologies.

⁵ The monoline credit card company CapitalOne is an example of a firm that experienced remarkable growth in the 1990s by more efficiently using demographic information on borrowers.

⁶ We track default outcomes for all originated loans for up to three years following origination, meaning that we follow the 2013 cohort up to 2016.

⁷ We also employ the eXtreme Gradient Boosting (XGBoost) model (Chen and Guestrin (2016)), which delivers similar results to the RF. For brevity, therefore, we focus on describing the results from the RF model here, and we provide details on XGBoost in the Internet Appendix, which is available in the online version of the article on *The Journal of Finance* website.

logistic models. We also find that predicted default propensities across race and ethnic groups look very different under the more sophisticated technology than under the simple technology. In particular, while a large fraction of borrowers who belong to the majority group (e.g., White non-Hispanic) “win,” that is, experience lower estimated default propensities under the machine learning technology than the less sophisticated Logit technology, these benefits do not accrue to some minority race and ethnic groups (e.g., Black and Hispanic borrowers) to the same degree. We show that these inferences are robust to numerous changes to the set of covariates, the sample used for estimation, and the estimation approach.

We propose simple empirical measures to bound the extent to which flexibility or triangulation is responsible for these results by comparing the performance of the naïve and sophisticated statistical models when race and ethnicity are included versus withheld from the information set used to predict default. While we find that both flexibility and triangulation are important, in our empirical application, the majority of the predictive accuracy gains from the more sophisticated machine learning model can be attributed to the increased flexibility of the model, with at most 30% of the gains attributable to pure triangulation. These findings suggest that simply prohibiting the use of certain variables as predictors of default propensity will likely become increasingly ineffective as technology improves.⁸ For one, such regulations will confront the difficulty of prohibiting triangulation in the face of increasingly complicated attempts to model the joint distribution of outcomes, permissible variables, and restricted characteristics.⁹ In addition, such regulations cannot protect minorities against the greater flexibility conferred by the new technology.

How might these changes in predicted default propensities across race and ethnic groups map into actual outcomes, that is, into the propensity that different groups of borrowers will be granted mortgages and the interest rates that they will be asked to pay when granted mortgages? In a first attempt to address these questions, we embed the statistical models in a simple equilibrium model of credit provision assuming a competitive credit market in which rational lenders compete to issue loans. To evaluate magnitudes, we assume that lenders are subject to a constraint arising from the availability of statistical prediction technology.¹⁰ We then compute counterfactual equilibria associated

⁸ In practice, compliance with the letter of the law has usually been interpreted to mean that differentiation between households using “excluded” characteristics such as race or gender is prohibited (e.g., Ladd (1998)).

⁹ We note that the machine learning models are better able than the logistic models to predict race using borrower information such as FICO score and income. This is reminiscent of recent work in the computer science literature that shows that anonymizing data is ineffective if sufficiently granular data on characteristics about individual entities are available (e.g., Narayanan and Shmatikov (2008)).

¹⁰ We consider a model in which lenders bear the credit risk on mortgage loans (which is the key driver of their accept/reject and pricing decisions) and are in Bertrand competition with one another. In contrast, the U.S. mortgage market over our sample period is one in which the vast

with each statistical technology and compare the resulting equilibrium outcomes to evaluate comparative statics on outcomes across groups.

In this simple analysis of counterfactuals arising under different technologies, we face a number of obvious challenges to identification. These challenges are due to the fact that the data that we use to estimate the default models were not randomly generated, but rather, a consequence of interactions between borrowers and lenders who may have had access to additional information when making their decisions. We attempt to address these challenges by changing the estimation sample and attempting to de-bias our estimates, as we describe later in the paper. We simply caveat here that the results of our elementary computations should not be viewed as a precise prediction, but as a useful first step toward assessing magnitudes.

We find that the machine learning model is predicted to provide a slightly larger number of borrowers access to credit and to marginally reduce disparity in acceptance rates (i.e., the extensive margin) across race and ethnic groups in the borrower population. The story is different, however, for the intensive margin—the cross-group disparity in equilibrium rates increases under the machine learning model relative to the less sophisticated logistic regression models. This result is accompanied by a substantial increase in within-group dispersion in equilibrium interest rates as technology improves. This increase is virtually double the magnitude for Black and White Hispanic borrowers under the machine learning model than for White non-Hispanic borrowers, that is, Black and Hispanic borrowers receive very different rates (within group) under the machine learning technology. If we focus on the (large) subset of borrowers who are predicted to be offered a mortgage under both technologies, this increase in rate dispersion represents a significantly larger penalty for risk-averse minority borrowers from an ex-ante perspective.

Overall, the picture is mixed. On the one hand, the machine learning model predicts default more accurately than the more primitive technologies. What's more, it appears to provide credit to a slightly larger fraction of mortgage borrowers and to slightly reduce cross-group dispersion in acceptance rates. However, the main effects of the improved technology are to increase rate dispersion both across and within race groups, especially for Black and Hispanic borrowers.

Our focus in this paper is on the distributional impacts of changes in technology rather than on explicit taste-based discrimination (Becker (1971)) or “redlining,” which seeks to use geographical information to indirectly differentiate on the basis of excluded characteristics and is proscribed.¹¹ That said, our

majority of loans are insured by government-backed entities that also set underwriting criteria and influence pricing. Our exercise can be viewed as an attempt to map the changes in default probabilities that we find on credit provision along the intensive and extensive margins, which is of interest whether new prediction technology is used by private lenders or by a centralized entity changing its approach to setting underwriting criteria.

¹¹ Bartlett et al. (2021) empirically study whether “FinTech” mortgage lenders in the United States appear to discriminate more across racial groups. Buchak et al. (2018) and Fuster et al. (2019) study other aspects of FinTech lending in the U.S. mortgage market.

exercise is similar in spirit to this work in the sense that we also seek a clearer understanding of the sources of inequality in household financial markets.¹² Our work is also connected more broadly to theories of statistical discrimination,¹³ although we do not model lenders as explicitly having access to racial and ethnic information when estimating borrowers' default propensities.

The organization of the paper is as follows. Section I sets up a basic theoretical framework to understand how improvements in statistical technology can affect different groups of households in credit markets, and describes how nonlinear technologies relate to the two sources of unequal effects (i.e., flexibility and triangulation). Section II discusses the U.S. mortgage data that we use in our study. Section III introduces the default forecasting models that we employ on these data, describes how predicted default probabilities vary across groups, and computes measures of flexibility and triangulation in the data. Section IV sets up our simple equilibrium model of credit provision under different technologies and discusses how changes in default predictions affect both the intensive and the extensive margins of credit provision. Section V concludes. An extensive Internet Appendix contains proofs, numerous auxiliary analyses, and robustness checks.

I. A Simple Conceptual Framework

Consider a lender predicting the probability of default, $y \in [0, 1]$, of a loan using a vector x of observable borrower characteristics (e.g., income, credit score) and contract terms (e.g., loan size). Each borrower is characterized by their x vector, as well as by their group membership g (e.g., their race). The lender is not permitted to include g in prediction.

The lender wishes to find a function $\hat{y} = \hat{P}(x)$ that maps the observable characteristics x into a predicted y . We represent the statistical technology that the lender can use to find this function by $\mathcal{M}$, which comprises a class of possible functions that can be chosen. We say that a statistical technology $\mathcal{M}_2$ is *better than* $\mathcal{M}_1$ if it gives the lender a larger set of functional options, that is, $\mathcal{M}_1 \subset \mathcal{M}_2$.

We view this definition as capturing the main conceptual differences between traditional statistical technology and machine learning. Machine learning techniques such as tree-based models and neural networks can employ a wider range of functional forms $\hat{P}(x)$ in prediction, relative to traditional approaches (e.g., Logit) that rely primarily on linear functions of x .¹⁴ Note that

¹² These issues have been a major focus of work on mortgages and housing. See, for example, Berkovec et al. (1994, 1998), Ladd (1998), Ross and Yinger (2002), Ghent, Hernández-Murillo, and Owyang (2014), Bayer, Ferreira, and Ross (2018), or Bhutta and Hizmo (2020). In insurance markets, see, for example, Einav and Finkelstein (2011), Chetty and Finkelstein (2013), Bundorf, Levin, and Mahoney (2012), and Geruso (2016). Also related, Pope and Sydnor (2011) consider profiling in unemployment benefits use.

¹³ See Fang and Moro (2010) for an excellent survey, as well as classic references on the topic, such as Phelps (1972) and Arrow (1973).

¹⁴ For simplicity, our theoretical exposition focuses on the case in which statistical technologies are exactly nested with $\mathcal{M}_1 \subset \mathcal{M}_2$. In practice, the classes of functional forms we consider in this

we study the distributional consequences of innovation in statistical technologies given a *fixed* set of observable variables x ; we do not consider the effects of expanding this set, say, by using borrowers' "digital footprints" (e.g., Berg et al. (2020)).

The standard goal of statistical learning is to find predictive functions $\hat{P}(x)$ that minimize a statistical loss function. For concreteness, we focus on MSE loss in this section. We define the predictive function minimizing this loss, subject to the constraint imposed by the available statistical technology, as follows:

$$\hat{P}(x|\mathcal{M}) = \arg \min_f E[(f(x) - y)^2] \text{ subject to } f \in \mathcal{M}. \quad (1)$$

It is important to note that the functions $\hat{P}(x|\mathcal{M}_j)$ are not directly observed in practice, because the statistician does not know the true joint distribution of x and y and hence cannot compute the expectation in (1). However, the fundamental results of statistical learning theory show that predictions obtained by empirical risk minimization converge to these "ideal" loss-minimizing predictors in large samples.¹⁵ Additional machine learning techniques such as regularization allow for faster convergence to the ideal because they discipline overfitting in finite samples.

Given this setup, to derive a large-sample approximation of the consequences of the change in technology, we compare the ideal predictor in $\mathcal{M}_1$ to that obtained in the broader class $\mathcal{M}_2$. We find that improvements in statistical technology lead to predictions that are more disperse across borrowers.

LEMMA 1: *Assume that $\mathcal{M}_2$ is a better statistical technology than $\mathcal{M}_1$, with $\mathcal{M}_1 \subset \mathcal{M}_2$. Then, in a population of borrowers, $\hat{P}(x|\mathcal{M}_2)$ is a mean-preserving spread of $\hat{P}(x|\mathcal{M}_1)$.*

The proof is provided in the Appendix.¹⁶ The result is intuitive: by definition, improvements in technology yield predictions with an MSE at least as small as that from preexisting predictions. These new predictions $\hat{y}$ track true y more closely, and will hence be more disperse on average.¹⁷

paper are nested only in an approximate sense. Our empirical application uses tree-based models (Random Forest, XGBoost), which work by combining simple indicator functions. While these can never exactly replicate a smooth functional form such as Logit, techniques such as Random Forests have sufficient flexibility to closely approximate functional forms ranging from those that are simple and linear to those that are significantly more complex ones. Furthermore, we note that traditional models such as Logit can (and, in our empirical application, do), of course, incorporate nonlinear terms and interactions, but unlike in machine learning models, these need to be prespecified and are typically used rather sparingly to avoid overfitting.

¹⁵ Empirical risk minimization, for a given class of functions $\mathcal{M}$, refers to finding the predictive function $P(x) \in \mathcal{M}$ that minimizes *realized* MSE in a training sample of data. See, for example, Vapnik (1999) and Friedman, Hastie, and Tibshirani (2001) for an exposition of statistical learning theory.

¹⁶ The proof imposes the additional technical condition that both $\mathcal{M}_1$ and $\mathcal{M}_2$ are closed subspaces of the space $\mathcal{L}^2$ of square-integrable functions of x .

¹⁷ The fact that the spread is mean-preserving follows because ideal predictors that minimize the MSE will be unbiased regardless of technology. This is not necessarily true of predictions

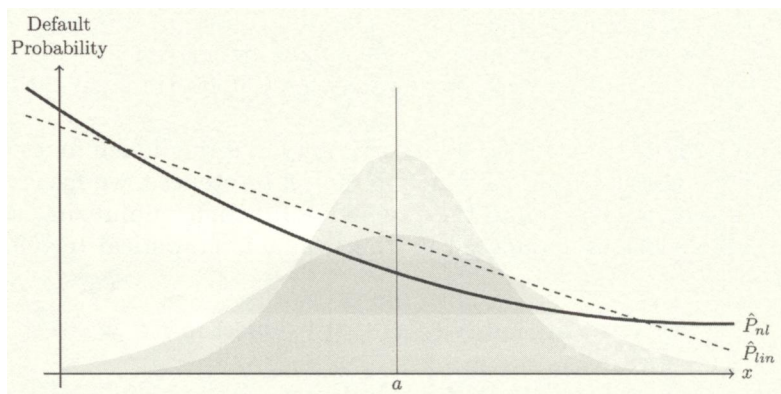


Figure 1. Unequal effects of better technology. (Color figure can be viewed at wileyonlinelibrary.com)

Lemma 1 shows that better technology shifts weight from average predicted default probabilities to more extreme values. As a result, some borrowers with characteristics x will be treated as less (more) risky under the new technology and experience better (worse) credit market conditions. Put differently, there will be both winners and losers as better technology becomes available in credit markets, motivating the distributional concerns at the heart of our analysis. However, this analysis does not provide guidance on the specific groups g of borrowers who will be better or worse off, a matter that we return to below.

Figure 1 illustrates possible group-specific effects when borrowers have only one observable characteristic x . There are two groups of borrowers, Blue and Red, with the same mean $x = a$ but different variances of x ; the two bell curves show these group-specific characteristics distributions. In this example, the traditional statistical technology consists of linear predictive functions, $\hat{P}_{lin}(x)$, while the more sophisticated statistical technology $\hat{P}_{nl}(x)$ is convex quadratic in x . We have $\hat{P}_{nl}(x) > \hat{P}_{lin}(x)$ when x is far from its mean a in either direction. It follows that Blue borrowers tend to be adversely affected by the new technology, as their characteristics x are more variable and hence more likely to lie in the tails of the distribution, which are penalized by the nonlinear technology.

This intuition about the factors determining winners and losers generalizes beyond the convex quadratic example, which we use simply for illustrative purposes. More generally, the effect of introducing a more sophisticated technology depends on two factors, namely, the higher order moments of borrower

achieved in finite samples, where machine learning techniques trade-off increases in bias against reductions the variance of out-of-sample forecast (see, for example, James et al. (2013)). However, these biased predictions still converge to MSE-minimizing predictions as the data set grows large. As a result, the properties discussed here are (once again) approximately informative about the properties of regularized estimators, as long as algorithms are fit on sufficiently large data sets (in our case, $N \approx 10$ million).

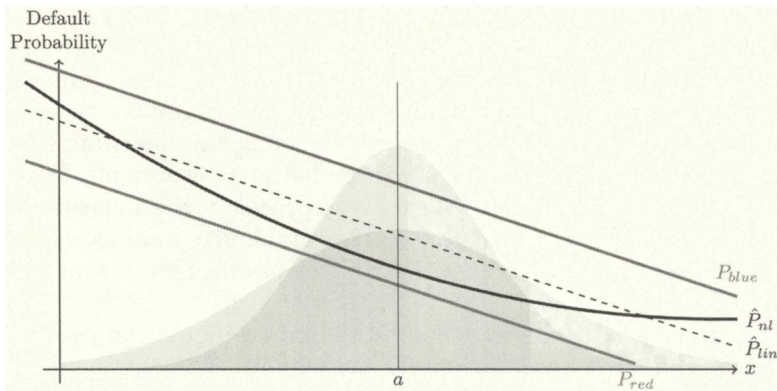


Figure 2. Triangulation. (Color figure can be viewed at wileyonlinelibrary.com)

characteristics in each group and the higher order derivatives of predictions under sophisticated technology.¹⁸

What are the underlying sources of unequal effects? We consider two widely discussed possibilities, as mentioned earlier in the Introduction. One is that the unequal effects across groups could be driven by the *flexibility* of the new technology. If the true function connecting x and y is nonlinear while g does not affect y , then the underlying source of the unequal effects is the ability of the new technology to capture this nonlinear structural relationship. Another possibility is that more sophisticated technology can *triangulate* group identity. Intuitively, in this case, the more sophisticated technology uses nonlinear functions of x to more effectively proxy for the relationship between the omitted variable g and y , thereby resulting in unequal effects under the new technology.¹⁹

Figure 2 provides an example where unequal effects are generated exclusively by triangulation. Here, true default risk is assumed to be a linear function of x in each group, and higher for the Blue group ($P_{blue} > P_{red}$), while the group-conditional distributions of x are the same as in Figure 1. The linear prediction $\hat{P}_{lin}(x)$ in this case will simply equal the population-weighted average of the true group-specific default probabilities (i.e., the dashed straight line in the figure). In contrast, the nonlinear technology penalizes the Blue group—since extreme realizations of x are more likely to come from Blue borrowers,

¹⁸ Lemma IA.1 in the Internet Appendix makes this point formally in the context of predictions that are smooth functions of a single characteristic x .

¹⁹ As discussed in the literature, g could capture the effects of unobservables such as access to informal safety nets (e.g., ability to borrow from family or friends), differential treatment in the labor market, access to other sources of formal credit, or indeed other sources of unobserved income or wealth that affect y . Our addition to this debate is that machine learning would yield differential predictions across groups because it better proxies the predictive power of omitted g for y by using nonlinear combinations of x .

the technology assigns higher predicted default probabilities to more extreme realizations of x .²⁰

These examples highlight two points that guide our empirical analysis. First, Figure 1 shows that the group-specific effects are ambiguous a priori. Without knowing the precise nature of nonlinearity in machine learning predictions, one cannot anticipate which groups will be better or worse off. For example, a concave quadratic function would deliver precisely the opposite effects as the convex quadratic that we posited.²¹ This ambiguity implies that one must inspect the data to understand the distributional effects of machine learning, as we do in the next section.

Second, Figure 2 suggests that flexibility and triangulation can deliver unequal effects that are observationally equivalent. The distinction between them is important, however, from a normative perspective, as the two scenarios would result in a very different set of conversations. For instance, triangulation might lead us to consider alternative regulations that are fit for purpose when lenders use highly nonlinear functions, such as the approaches proposed in Ross and Yinger (2002) or Pope and Sydnor (2011), whereas flexibility might instead lead us toward discussing the underlying sources of cross-group differences in the distributions of observable characteristics. In Section III, we define empirical measures of flexibility and triangulation in an attempt to bound the extent to which these two mechanisms drive unequal effects observed in the data.

II. U.S. Mortgage Data

To study how these issues may play out in reality, we use high-quality administrative data on the U.S. mortgage market, which results from merging two loan-level data sets: (i) data collected under the Home Mortgage Disclosure Act (HMDA) and (ii) the McDashTM mortgage servicing data set, which is owned and licensed by Black Knight.

HMDA data have traditionally been the primary data set used to study unequal access to mortgage finance by loan applicants of different races, ethnicities, or genders—indeed, “identifying possible discriminatory lending patterns” was one of the main purposes in establishing HMDA in 1975.²² HMDA reporting is required of all lenders above a certain size threshold who are active in metropolitan areas, and the HMDA data are thought to cover 90% or more of all first-lien mortgage originations in the United States (e.g., National Mortgage Database (2017), Dell’Ariccia, Igan, and Laeven (2012)).

²⁰ In the Internet Appendix, we provide a mathematical derivation of the pattern shown in this figure.

²¹ That is, new technology could allow a lender to identify good credit risks within a minority group previously assigned uniformly high predicted default rates under the old technology, thus *reducing* inequality across groups. Anecdotally, credit card company CapitalOne more efficiently used demographic information and expanded lending in such a manner over the 1994 to 2004 period. See, for example, Wheatley (2001).

²² See <https://www.ffiec.gov/hmda/history.htm>.

HMDA lacks a number of key pieces of information that we need for our analysis. First, loans in this data set are observed only at origination, so it is impossible to know whether a borrower in the HMDA data set ultimately defaulted on an originated loan. Moreover, a number of borrower characteristics useful for predicting default are missing from the HMDA data, such as the credit score (FICO), the loan-to-value ratio (LTV), the term of the issued loan, and loan cost (this information is only reported for “high cost” loans).²³

The McDashTM data set from Black Knight contains much more information on the loan contract and borrower characteristics, including mortgage interest rates. Of course, these data are only available for originated loans, which the data set follows over time. The data set also contains a monthly indicator of a loan’s delinquency status, which makes it one of the primary data sets that researchers use to study mortgage default (e.g., Elul et al. (2010), Foote et al. (2010), Ghent and Kudlyak (2011)).

A matched data set of HMDA and McDash loans is made centrally available to users within the Federal Reserve System. The match is done by origination date, origination amount, property zipcode, lien type, loan purpose (i.e., purchase or refinance), loan type (e.g., conventional or FHA), and occupancy type. We only retain loans that can be uniquely matched between HMDA and McDash; we discuss how this affects our sample size below.

Our entire data set runs from 2009 to 2016. We use these data to estimate three-year probabilities of delinquency (i.e., three or more missed payments, also known as “90-day delinquency”) on all loans originated between 2009 and 2013.²⁴ We thus focus on loans originated after the end of the housing boom, which (unlike earlier vintages) did not experience severe declines in house prices. Indeed, most borrowers in our data experienced positive house price growth throughout the sample period. This means that delinquency is likely driven largely by idiosyncratic borrower shocks rather than macro shocks, mapping more closely to our theoretical discussion.

For origination vintages from 2009 to 2013, our HMDA-McDash data set corresponds to 45% of all loans in HMDA. This fraction is driven by the coverage of McDash (corresponding to 73% of HMDA originations over this period) and the share of these McDash loans that can be uniquely matched to HMDA loans (just over 60%). For our analysis, we impose additional sample restrictions. We only retain conventional (nongovernment issued) fixed-rate first-lien mortgages on single-family homes and condo units with an original loan term of 10, 15, 20, or 30 years. Furthermore, we only keep loans with original LTV between 20% and 100%, a loan amount of 1 million U.S.\$ or less, and borrower income of 500,000 U.S.\$ or less. We also drop observations where the occupancy type

²³ Bhutta and Ringo (2014) and Bayer, Ferreira, and Ross (2018) merge HMDA data with information from credit reports and deeds records in their studies of racial and ethnic disparities in the incidence of high-cost mortgages. Since the 2018 reporting year, additional information has been collected under HMDA; see http://files.consumerfinance.gov/f/201510_cfpb_hmda-summary-of-reportable-data.pdf for details.

²⁴ We estimate three-year probabilities of delinquency to ensure that censoring of defaults affects all vintages similarly for comparability.

is marked as unknown. Finally, we require that the loans reported in McDash have data beginning no more than six months after origination, which is the case for the majority (about 83%) of the loans in McDash originated over our sample period. This requirement that loans not be excessively “seasoned” before data reporting begins attempts to mitigate selection bias associated with late reporting.

There are 42.2 million originated mortgages on one to four family properties (including manufactured homes) in the 2009 to 2013 HMDA data. The matched HMDA-McDash sample imposing the seasoning restriction contains 16.84 million loans, of which 72% are conventional loans. After imposing all of our remaining data filters on this sample, we end up with 9.37 million loans. For all of these loans, we observe whether they enter serious delinquency at any point over the first three years of their life—this occurs for 0.74% of these loans.

HMDA contains separate identifiers for race and ethnicity; we focus primarily on race, with one important exception. For White borrowers, we additionally distinguish between Hispanic/Latino White borrowers and non-Hispanic White borrowers.²⁵ The number of borrowers in each group, along with descriptive statistics for key observable variables, are reported in Table I.

The table shows that there are clear differences between the (higher) average and median FICO scores, income levels, and loan amounts for White non-Hispanic and Asian borrowers relative to Black and White Hispanic borrowers. Moreover, the table shows that average default rates (as well as interest rates and the spreads at origination over average interest rates, known as “SATO”) are higher for the Black and White Hispanic borrowers. The latter two groups also exhibit substantially higher variance in FICO scores than the White Non-Hispanic group. Intuitively, such differences in characteristics make these minority populations look different from the “representative” borrower discussed in the single-characteristic model of default probabilities in Section I. Depending on the shape of the functions under the new statistical technology, these differences will be either penalized or rewarded (in terms of estimated default probabilities) under the new technology relative to the old.

Even though the sample we use for our analysis covers a sizable portion of U.S. mortgage originations during the sample period, one may still be concerned that our sample is not fully representative. In the Internet Appendix, we show that at least based on key characteristics like income and loan amount, there is no evidence that the distributions of these variables across and within groups are not representative of the market (as measured in

²⁵ The different race codes in HMDA are: (1) American Indian or Alaska Native; (2) Asian; (3) Black or African American; (4) Native Hawaiian or Other Pacific Islander; (5) White; (6) information not provided by applicant in mail, Internet, or telephone application; and (7) not applicable. We combine groups (1) and (4) due to the low number of borrowers in each of these categories; we also combine groups (6) and (7) and refer to the combined category as “Unknown.” (We check robustness to dropping this category prior to estimation.) Ethnicity codes are: Hispanic or Latino; Not Hispanic or Latino; Information not provided by applicant in mail, Internet, or telephone application; and Not applicable. We classify a borrower as Hispanic only in the first case, and we make the distinction only for White borrowers.

Table I
Descriptive Statistics, 2009 to 2013 Originations

This table reports summary statistics for different demographic groups. Income and loan amount are measured in thousands of USD. SATO stands for “spread at origination” and is defined as the difference between a loan’s interest rate and the average interest rate of loans originated in the same calendar quarter. Default is defined as being 90 days or more delinquent at some point over the first three years after origination. Data source: HMDA-McDash matched data set of fixed-rate mortgages originated over 2009 to 2013.

Group		FICO	Income	LoanAmt	Rate (%)	SATO (%)	Default (%)
Asian (N=574,812)	Mean	764	122	277	4.24	−0.07	0.42
	Median	775	105	251	4.25	−0.05	0.00
	SD	40	74	149	0.71	0.45	6.49
Black (N=235,673)	Mean	735	91	173	4.42	0.11	1.88
	Median	744	76	146	4.50	0.12	0.00
	SD	58	61	109	0.71	0.48	13.57
White Hispanic (N= 381,702)	Mean	746	90	187	4.36	0.07	0.99
	Median	757	73	159	4.38	0.07	0.00
	SD	52	63	115	0.71	0.47	9.91
White Non-Hispanic (N=7,134,038)	Mean	761	110	208	4.33	−0.00	0.71
	Median	774	92	178	4.38	0.02	0.00
	SD	45	73	126	0.69	0.44	8.37
Native Am., Alaskan, Hawaii/Pac Isl (N=59,450)	Mean	749	97	204	4.39	0.04	1.12
	Median	761	82	175	4.45	0.04	0.00
	SD	51	65	123	0.70	0.46	10.52
Unknown (N=984,310)	Mean	760	119	229	4.38	0.00	0.79
	Median	773	100	197	4.50	0.02	0.00
	SD	46	78	141	0.68	0.44	8.85

HMDA). We also verify the robustness of our empirical results to a number of sample perturbations, as we describe below.

It is worth noting another point regarding our data and the U.S. mortgage market more broadly. The vast majority of loans in the sample (over 90%) end up securitized by the government-sponsored enterprises (GSEs) Fannie Mae or Freddie Mac, which insure investors in the resulting mortgage-backed securities against the credit risk on the loans. Furthermore, these firms provide lenders with underwriting criteria that dictate whether loans are eligible for securitization and influence the pricing of loans.²⁶ As a result, the lenders retain originated loans in portfolio (i.e., on balance sheet) and thus directly bear the risk of default for less than 10% of the loans in our sample.

As we discuss below, when we study counterfactual equilibria associated with new statistical technologies, this feature of the market makes it less likely that there is selection on unobservables by lenders originating GSE-securitized

²⁶ For instance, in addition to their flat “guarantee fee” (i.e., insurance premium), the GSEs charge so-called “loan-level price adjustments” that depend on borrower FICO score, LTV ratio, and some other loan characteristics.

loans, which is important for identification. Nevertheless, in this section, we estimate default probabilities using both GSE-securitized loans and portfolio loans, in the interest of learning about default probabilities using as much data as possible, as we believe a profit-maximizing lender would also seek to do.²⁷

In the next section, we estimate increasingly sophisticated statistical models to predict default in the mortgage data set. We then evaluate how the predicted probabilities of default (PDs) generated by these models vary across race-based groups in the population of mortgage borrowers.

III. Estimating Probabilities of Default Using Different Statistical Technologies

In this section, we describe the different prediction methods that we employ to estimate default probabilities for originated mortgages in the our data set. In our description of the estimation techniques, we denote observable characteristics by x , the loan interest rate by R , and the conditional probability of default by $P(x, R) = \Pr(\text{Default}|x, R)$.²⁸ We subsequently use our estimated default probabilities to understand the impact of different statistical technologies on mortgage lending.

In the remainder of this section, given concerns of unobserved private information in issued loan interest rates R , we estimate $\hat{P}(x)$, that is, we do not include the loan rate at origination in the set of covariates. In practice, as we demonstrate below, our inferences are not much affected by the inclusion or exclusion of R when estimating default probabilities, and we attempt to more cleanly estimate $\hat{P}(x, R)$ for a simple back-of-the-envelope calculation of potential economic magnitudes on interest rates and loan granting decisions.

We also note that we restrict attention in this paper to the prediction of default probabilities. In practice, final outcomes such as interest rates should reflect not only default probabilities, but also factors such as borrowers' prepayment propensities, which may also have a group-specific component.²⁹ While this is a shortcoming of our approach, we nevertheless abstract from this issue in our analysis, for three main reasons. First, unlike default, prepayment

²⁷ One set of lenders that may have been using more sophisticated models during our sample period are "FinTech" lenders like Quicken Loans, which gained market share over the sample period. In our matched sample, we do not have lender identifiers (due to data restrictions), so we cannot directly study whether those lenders appear to assess and price risk differently. However, based on the list of FinTech lenders from Buchak et al. (2018) and the full HMDA sample, we note that the market share of these FinTech lenders was very low over the first three years of our sample (2% to 3% of all originated loans over the 2009 to 2011 period), before roughly doubling in 2012 and 2013. In our robustness checks, we show that our results are very similar if we restrict the sample to 2009 to 2011, making it unlikely that the patterns in the data that drive our results were due to FinTech lenders.

²⁸ We do not directly estimate lifetime PDs (which are the object of interest in our models in Sections I and IV), but rather three-year PDs. In the Internet Appendix, we discuss the industry-standard assumptions that we use to convert estimated three-year probabilities into lifetime PDs.

²⁹ Borrowers in the U.S. mortgage market, unlike in almost all other countries, generally have the option to prepay their loan at any time, without compensating the lender for lost income.

has ambiguous effects on the lender: the lender benefits (suffers) from faster prepayment when the rate on a loan is below (above) the prevailing market rate. To the extent that new loans are issued at par, as we later assume in our model (after accounting for credit risk), prepayment propensities do not have first-order effects on loan values.³⁰ Second, for our purposes in this paper, differences in prepayment behavior must manifest themselves systematically across groups to affect our inferences—and any such differences will have ambiguous effects depending on whether some groups systematically prepay quicker or slower than others conditional on how the market mortgage rate compares to the rate on their current loan.³¹ Third, to get a sense of how estimated differences in prepayment behavior would affect equilibrium interest rates across groups (in the spirit of our calculations in Section IV) would require rather complicated machinery (e.g., simulations from calibrated interest rate models) that beyond the scope of this paper.

We now describe the estimation approaches that we use to contrast traditional and more sophisticated prediction technologies. First, we implement two Logit models to approximate the “standard” prediction technology typically used by both researchers and industry practitioners (e.g., Demyanyk and Van Hemert (2011), Elul et al. (2010)). Second, to shed light on how more sophisticated prediction technologies affect outcomes across groups, we estimate a tree-based model and augment it using a number of techniques commonly employed in machine learning applications. More specifically, the main machine learning model that we consider is a RF model (Breiman (2001)); we use cross-validation and calibration to augment the performance of this model.³²

A. Logit Models

We begin by estimating two variants of a standard Logit model. These models find widespread use in default forecasting applications, with a link function such that

$$\log \left(\frac{g(x)}{1 - g(x)} \right) = x' \beta. \quad (2)$$

³⁰ See Gabaix, Krishnamurthy, and Vigneron (2007) for a simple model illustrating these points. Boyarchenko, Fuster, and Lucca (2019) show empirically that prepayment risk premia are close to zero for mortgage-backed securities with near-the-money prepayment options, as is the case for newly issued loans. Consistent with this result, in practice, prepayment modeling is most important for the valuation of older existing loans, which may have mortgage rates well above or below current market rates.

³¹ See, for instance, Keys, Pope, and Pope (2016) and Andersen et al. (2020) for recent studies of heterogeneity in mortgage refinancing behavior.

³² While many different techniques can be classified as belonging (or not) to the class of “machine learning” models, we simply seek to shed light on the effects of access to a flexible non-linear technology unencumbered by concerns of overfitting. This guides the comparison we draw between more traditional Logit-based approaches and the RF implemented and tuned with cross-validation and calibration. We also employ the eXtreme Gradient Boosting (XGBoost) model (Chen and Guestrin (2016)), which delivers very similar results to the RF—we describe this alternative model in the Internet Appendix.

Table II
Variables

This table lists the variables used in the main models. Section III.E considers additional specifications. Data source: HMDA-McDash matched data set of conventional fixed-rate mortgages.

Logit	Nonlinear Logit
Applicant Income (linear)	Applicant Income (25k bins, from 0 to 500k)
LTV Ratio (linear)	LTV Ratio (five-point bins, from 20% to 100%; separate dummy for LTV=80%)
FICO (linear)	FICO (20-point bins, from 600 to 850; separate dummy for FICO<600)
(with dummy variables for missing values)	
Common Covariates	
Origination Amount (linear and log)	
Documentation Type (dummies for full/low/no/unknown documentation)	
Occupancy Type (dummies for vacation/investment property)	
Jumbo Loan (dummy)	
Coapplicant Present (dummy)	
Loan Purpose (dummies for purchase, refinance, home improvement)	
Loan Term (dummies for 10, 15, 20, and 30 year terms)	
Funding Source (dummies for portfolio, Fannie Mae, Freddie Mac, other)	
Mortgage Insurance (dummy)	
State (dummies)	
Year of Origination (dummies)	

We estimate the model in two ways, varying how the covariates in x enter the right-hand side. In the first model, all of the variables in x (listed in Table II) enter linearly, and we include dummies for origination year, documentation type, occupancy type, product type, investor type, loan purpose, coapplicant status, and an indicator for whether the mortgage is a “jumbo” loan (meaning the loan amount is too large for Fannie Mae or Freddie Mac to securitize the loan). In addition, we include the term of the mortgage and state fixed effects. We refer to this model simply as the “Logit” in what follows.

In the second type of Logit model, we allow for more flexible use of the information in the covariates in x , reflecting standard industry practice. In particular, we include the same dummies as in the first model, but instead of all continuous variables entering the model for the log-odds ratio linearly, we bin some of them to allow for the possibility of nonlinear relationships. In particular, we assign LTV to bins of 5% width ranging from 20% to 100%, along with an indicator for LTV equal to 80, as this is a frequent value in the data. For FICO, we use bins of 20-point width from 600 to 850 (the maximum). We assign all FICO values between 300 (the minimum) and 600 into a single bin, since there are only few observations with such low credit scores. Finally, we

bin income into 25,000 U.S.\$ width bins from 0 to 500,000 U.S.\$. We refer to the resulting model as the “Nonlinear Logit” (NL).³³

B. Tree-Based Models

As an alternative to the traditional models described above, we use machine learning models to estimate $\hat{P}(x)$. The term is quite broad, but essentially refers to the use of a range of techniques to “learn” the function f that can best predict a generic outcome variable y using underlying attributes x . Within the broad area of machine learning, settings such as ours in which the outcome variable is discrete (here, binary, as we are predicting default) are known as *classification* problems.

Several features differentiate machine learning approaches from more standard approaches. For one, the former tend to be nonparametric. Another difference is that machine learning approaches generally use computationally intensive techniques such as bootstrapping and cross-validation, which have experienced substantial growth in applied settings as computing power and the availability of large data sets have both increased.

While many statistical techniques and approaches can be characterized as machine learning, we focus here on a set of models that have been both successful and popular in prediction problems, and that are based on the use of simple decision trees. In particular, we employ the RF technique (Breiman (2001)). In essence, the RF is a nonparametric and nonlinear estimator that flexibly bins the covariates x in a manner that best predicts the outcome variable of interest. As this technique has been fairly widely used, we provide only a brief overview of the technique here.³⁴

The RF approach can be best understood in two parts. First, a simple decision tree is estimated by recursively splitting covariates (taken one at a time) from a set x to best identify regions of default y . To fix ideas, assume that there is a single covariate under consideration, namely, LTV. To build a (primitive) tree, we would begin by searching for the single LTV value that best separates defaulters from nondefaulters, that is, maximizes a criterion such as cross-entropy or the Gini coefficient in the outcome variable between the two resulting bins on either side of the selected value, thus ensuring default prediction purity of each bin (or “leaf” of the tree). The process then proceeds recursively within each such selected leaf.

When applied to a broad set of covariates, the process allows for the possibility of bins in each covariate as in the NL model described earlier, but rather than the lender prespecifying the bin-ends, the process is fully data-driven as the algorithm learns the best function on a *training* subset of the data set

³³ As we describe below, we check robustness to allowing for an even richer set of right-hand-side variables in the NL, adding interactions between FICO and LTV bins, and further interacting these bins with loan purpose, term, and documentation type. Our inferences are not greatly affected by these changes.

³⁴ For a more in-depth discussion of tree-based models applied to a default forecasting problem, see, for example, Khandani, Kim, and Lo (2010).

for subsequent evaluation on an omitted subset of out-of-sample *test* data. An even more important differentiating factor is that the process can flexibly identify *interactions* between covariates, that is, bins that identify regions defined by (possibly nonlinear functions of) multiple variables simultaneously, rather than restricting the covariates to enter additively into the link function, or specifying the variable interactions upfront, as with the NL model.

The simple decision tree model is intuitive, and fits the data extremely well in-sample, that is, has low bias in the language of machine learning. However, it is typically quite bad at predicting out of sample, with extremely high variance on data sets that it has not been trained on; generally a result of overfitting on the training sample. To address this issue, the second step in the RF model is to implement (b)ootstrap (ag)gregation, or “bagging,” techniques. This approach attempts to reduce the variance of out-of-sample predictions without introducing additional bias. It does so in two ways. First, rather than fit a single decision tree, it fits many (500 in our application), with each tree fitted to a bootstrapped sample (i.e., sampled with replacement) from the original data set. Second, at each point at which a new split on a covariate is required, the covariate in question must come from a randomly selected subset of covariates. The final step when applying the model is to take the modal prediction across all trees when applied to a new (i.e., unseen/out-of-sample) observation of covariates x . The two approaches, that is, bootstrapping the data and randomly selecting a subset of covariates at each split, effectively decorrelate the predictions of the individual trees, providing greater independence across predictions. This reduces the variance in the predictions without much increase in bias (for textbook treatments, see, for example, Hastie, Tibshirani, and Friedman (2009), and James et al. (2013)).

A final note on cross-validation is in order here. Several (tuning) parameters must be chosen in the estimation of the RF model. Common parameters of this nature include, for example, the maximum number of leaves that the model is allowed to have in total and the minimum number of data points needed in a leaf to proceed with another split. To ensure the best possible fit, a common approach is to cross-validate the choice of parameters using K -fold cross-validation. This involves randomly splitting the training sample into K folds or subsamples (in our case, we use $K = 3$).³⁵

For each data fold, we estimate the model using a given set of tuning parameters on the remaining folds of the data (i.e., the remaining two-thirds of the training data in our setting with $K = 3$). We then check the fit of the resulting model on the omitted K th data fold. The procedure is then repeated K times, with the performance of the selected set of tuning parameters averaged across the folds. The entire exercise is then repeated for each point in a grid of potential tuning parameter values. Finally, the set of parameters that maximize the

³⁵ The choice of the hyperparameter K involves a trade-off between computational speed and variance; with a smaller K , there will be more variance in our estimates of model fit, as we will have fewer observations to average over, while with larger K , there will be a tighter estimate at the cost of more models to fit. Given that our RF model is computationally costly to estimate with 500 trees, to balance these considerations, we choose $K = 3$ to select tuning parameters.

out-of-sample fit in the cross-validation exercise are chosen. In our application, we cross-validate over the minimum number of data points needed to split a leaf, and the minimum number of data points required on a leaf.³⁶ Our procedure selects a minimum number of observations to split of 200 and requires at least 100 observations in each leaf.

B.1. Translating Classifications into Probabilities

An important difference between the RF model and the Logit models is that the latter naturally produce estimates of the probability of default given x , while the RF model (and indeed, many machine learning models focused on generating “class labels”) is geared toward providing a binary classification, that is, given a set of covariates, the model will output whether or not the borrower is predicted to default. For many purposes, including credit evaluation, the *probability* of belonging to a class (i.e., the default probability) is also needed, to set interest rates, for example. We therefore need to convert predicted class labels into predicted loan default probabilities to serve as inputs into a model of lending decisions.

In tree-based models such as the RF model, we could estimate these probabilities by counting the fraction of predicted defaults in the training data set associated with the leaf into which a new borrower is classified. However, such estimates tend to be very noisy, as leaves are optimized for purity, and there are often sparse observations in any given leaf. A frequently used alternative in machine learning is to use an approach called “calibration,” in which noisy estimated probabilities are refined/smoothed by fitting a monotonic function to transform them (see, for example, Niculescu-Mizil and Caruana (2005)). Common transformations include running a logistic regression on these probabilities to connect them to the known default outcomes in the training data set (“sigmoid calibration”), and searching across the space of monotonic functions to find the best fit function connecting the noisy estimates with the true values (“isotonic regression calibration”).³⁷ We employ isotonic regression calibration to translate the predicted classifications into probability estimates, providing more details of this procedure in the Internet Appendix.

B.2. Estimation

As mentioned earlier, we first estimate both sets of models (the two Logit versions and the RF) on a subset of our full sample, which we refer to as the *training* set. We then evaluate the performance of the models on a *test* set,

³⁶ We define our grid from 2 to 500 in increments of 50 (i.e., 2, 50, 100, etc.) for the minimum number of data points needed to split (*min_samples_split*), and a grid from 1 to 250 in increments of 50 for the minimum number of data points in a leaf (*min_samples_leaf*).

³⁷ In practice, the best results are obtained by estimating the calibration function on a second “calibration training set” that is separate from the training data set on which the model is trained. The test data set is then the full data set less than the two training data sets. See, for example, Niculescu-Mizil and Caruana (2005). We use this approach in our empirical application.

which the models have not seen before. In particular, we use 70% of the sample to estimate and train the models, and 30% to test the models. When we sample, we randomly select across all loans, such that the training and test samples are chosen independent of any characteristics, including year of origination.

We also further split the training sample into two subcomponents: 70% of the training sample is used as a *model* sample, which we use to estimate the Logit and NL models and to train the RF model, and the remaining 30% of the training data comprises the *calibration* sample, which we use to estimate the isotonic regression to construct probabilities from the predicted RF class labels as described above. This ensures that both sets of models employ the same amount of data to estimate default probabilities.³⁸

C. Model Performance

We evaluate the performance of the different models on the test set in several ways. We first plot receiver-operating characteristic (ROC) curves, which show the variation in the true positive rate (TPR) and the false positive rate (FPR) as we vary the probability threshold for declaring an observation to be a default (e.g., >50% is customary in Logit). A popular metric used to summarize the information in the ROC curve is the area under the curve (AUC; see, for example, Bradley (1997)). Models for which AUC is higher are preferred, as these are models for which the ROC curve is closer to the northwest (higher TPR for any given level of FPR).³⁹

One drawback of the AUC is that it is less informative in data sets that are sparse in defaulters, since FPRs are naturally low in such data sets (see, for example, Davis and Goadrich (2006)). We therefore also compute the *Precision* of each classifier, $P(y = 1|\hat{y} = 1)$, and the *Recall* of each classifier as $P(\hat{y} = 1|y = 1)$,⁴⁰ and draw precision-recall curves that plot precision against recall for different probability thresholds. To summarize these precision-recall curves, we report the average precision score, which calculates the weighted mean of precision, with weights corresponding to the trade-off in recall.⁴¹

Two additional measures that we compute are the Brier Score and the R^2 . The Brier Score is calculated as the average squared prediction error. Since this measure captures total error in the model, a smaller number is better, unlike the other metrics. The Brier Score can be decomposed into three

³⁸ We estimate the RF model using Python's scikit-learn package, and the Logit models using Python's statsmodels package.

³⁹ The TPR is the fraction of true defaulters in the test set that are also (correctly) predicted to be defaulters, and the FPR is the fraction of true nondefaulters in the test set (incorrectly) predicted to be defaulters. An intuitive explanation of the AUC is that it captures the probability that a randomly picked defaulter will have been ranked more likely to default by the model than a randomly picked nondefaulter.

⁴⁰ Note that the *Recall* is equal to the TPR.

⁴¹ Specifically, average *Precision* = $\sum_n (R_n - R_{n-1})P_n$, where n denotes each point on the precision-recall curve, and R_n and P_n denote the recall and precision at point n .

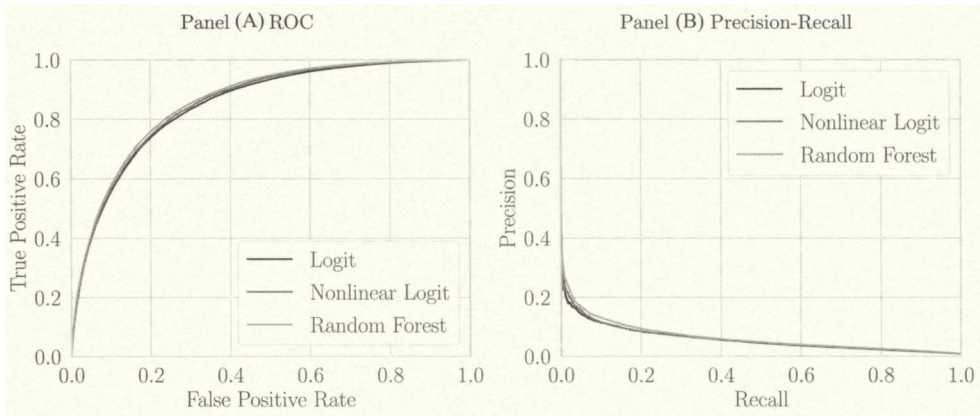


Figure 3. ROC and precision-recall curves. (Color figure can be viewed at wileyonlinelibrary.com)

components:

$$n^{-1} \sum_n (\hat{P}(x_i) - y_i)^2 = n^{-1} \underbrace{\sum_{k=1}^K n_k (\hat{y}_k - \bar{y}_k)^2}_{\text{Reliability}} - n^{-1} \underbrace{\sum_{k=1}^K n_k (\bar{y}_k - \bar{y})^2}_{\text{Resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{Uncertainty}}, \quad (3)$$

where the predicted values are grouped into K discrete bins, $\hat{y}_k$ is the predicted value within the k th bin, and $\bar{y}_k$ is the true mean value within the k th bin. Uncertainty measures an inherent feature of the outcomes in the prediction problem, reliability is a measure of the model's calibration, that is, the distance between the predicted probabilities and the true probabilities, and resolution is a measure of the spread of the predictions. Larger resolution is better, while smaller reliability implies a smaller overall error. In our application, overall uncertainty is 0.00735, and tends to dominate the overall value of the Brier Score. Finally, the R^2 is calculated as one minus the sum of squared residuals under the model, scaled by the sum of squared residuals from using the simple mean, with the usual interpretation as the percentage share of overall variance of the left-hand-side variable explained by a model.

Panels A and B of Figure 3 plot the ROC and precision-recall curves on the test data set for the three models that we consider. Both figures show that the RF model performs better than both versions of the Logit model. In Panel A, the TPR appears to be weakly greater for the RF model than for the traditional models for every level of FPR. In Panel B, the precision-recall curves, which are better suited for evaluating models on the kind of data set we consider (sparse in defaulters), show stronger gains for the RF model over the Logit models.

Columns (1), (4), (7), and (10) of Table III show that the RF model performs better than the Logit model, suggesting that the machine learning model more efficiently uses information in the training data set to generate more accurate predictions out of sample. The RF outperforms the NL model by 5.1% in terms

Table III
Performance of Different Statistical Technologies Predicting Default

This table shows performance metrics of different models. For ROC AUC, Precision Score, and R^2 , higher numbers indicate higher predictive accuracy; for the Brier Score, lower numbers indicate higher accuracy. In columns (1), (4), (7), and (10), race indicators are not included in the prediction models; in the other columns, they are included. In columns (2), (5), (8), and (11), the Logit models include the race indicators as simple dummy variables, while in columns (3), (6), (9), and (12), separate Logit models are estimated for each race group (which is equivalent to fully interacting the race indicators with all other variables).

Model	ROC AUC			Precision Score			Brier Score $\times 100$			R^2		
	(1) No Race	(2) Race	(3) Race Int.	(4) No Race	(5) Race	(6) Race Int.	(7) No Race	(8) Race	(9) Race Int.	(10) No Race	(11) Race	(12) Race Int.
Logit	0.8486	0.8491	0.8499	0.0579	0.0582	0.0589	0.7181	0.7179	0.7173	0.0233	0.0234	0.0243
Nonlinear Logit	0.8537	0.8541	0.8543	0.0590	0.0593	0.0600	0.7149	0.7148	0.7147	0.0275	0.0277	0.0279
Random Forest	0.8602	0.8610		0.0620	0.0622		0.7120	0.7118		0.0315		0.0318

of average precision, 0.8% in terms of AUC, and 14.3% in terms of R^2 . The Brier Score, as mentioned, is dominated by the overall uncertainty of the outcome (which is driven by the overall probability of default). After adjusting for this, the change from NL to RF is more substantial (14.3%), identical to the R^2 improvement. The decomposition reveals that the gains in reliability from switching to RF are large, with a reduction in reliability error of 94.3%, but at the cost of a small (3.2%) decline in resolution.⁴²

While these differences in performance metrics appear modest, we verify that they are indeed statistically significant using bootstrapping. To do so, we hold fixed the training and test splits, and resample observations with replacement from the original training data set to create 100 bootstrapped sample training data sets. We then reestimate all models on each bootstrapped training data set and evaluate them on the held-out test sample. Across all metrics, the RF shows better performance than the NL in 100% of these bootstrap samples.⁴³ Overall, we conclude with considerable statistical confidence that the machine learning models statistically significantly improve default prediction performance.

C.1. Model Performance with and without Race

Columns (2), (5), (8), and (11) of Table III simply add race dummies to the Logit models and show that this inclusion positively affects the performance of the traditional Logit models. In columns (3), (6), (9), and (12), we allow each variable in the Logit models to be *interacted* with the race dummies, which is equivalent to estimating separate Logits for each group. The performance increase from this change is even greater. The table also demonstrates how the more sophisticated machine learning model performs when it is allowed to use race as an explanatory variable—the model flexibly picks race and the interactions of race with the other included variables, so we simply show the single measure of performance from adding race to the set of features. We find that even this more sophisticated model benefits from the inclusion of race as an explanatory variable.

That said, it is worth noting that while all models benefit from the inclusion of race either as a dummy or when interacted with the other variables in the model, the improvement is quite small relative to that conferred by the increased sophistication of the model that is used. For example, the increase in R^2 obtained from moving to the RF model to the simple Logit dwarfs any improvement obtained from adding race as a variable to any of the models.

⁴² This result is consistent with Figure IA.2 in the Internet Appendix, where we see more spread in the predictions of the Logit model, but slightly worse calibration. When calculating the Brier Score decomposition, we use bins of size 0.0001 and restrict our predicted values in the model to less than 0.15 to avoid the influence of outliers.

⁴³ The histograms across bootstrapped data sets of the difference in these scores between the RF and the NL models are shown in Figure IA.6 in the Internet Appendix. A *t*-test based on the bootstrapped differences in performance statistics rejects the null hypothesis that the true mean difference in performance statistics (if we ran all possible training sets) is ≤ 0 at $p < 0.001$.

Table IV
Performance of Different Statistical Technologies Predicting Race

This table shows performance metrics of different models. For ROC AUC, Precision Score, and R^2 , higher numbers indicate higher predictive accuracy; for the Brier Score, lower numbers indicate higher accuracy.

Model	ROC AUC	Precision Score	Brier Score $\times 10$	R^2
Logit	0.7436	0.1837	0.5792	0.0607
Nonlinear Logit	0.7442	0.1857	0.5785	0.0619
Random Forest	0.7485	0.1983	0.5738	0.0694

Evaluating changes in the models’ predictive ability as a result of including race is interesting. In keeping with the spirit of the law prohibiting differentiation between borrowers on the basis of excluded characteristics such as race, assessments of borrower risk should be colorblind. The fact that race appears to augment performance for all models suggests that there is still some sense in which this restriction might be helpful. Importantly, even though the performance improvement magnitudes are small overall, this does not mean that the race indicators do not have significant effects for some groups—for instance, average default probabilities in the NL increase from 0.016 to 0.019 for Black borrowers when race indicators are added, while they decrease for Asian borrowers from 0.006 to 0.004.

To explore this issue further, we examine the extent to which the models can predict the ethnicity/race of borrowers using the same set of variables used to predict default (we do so by predicting an indicator for whether or not a borrower is Hispanic or Black). This exercise continues to reveal differences between the models—Table IV confirms that the RF outperforms the two other models, which have very similar scores, by 6.7% in terms of average precision, 0.6% in terms of AUC, 1.1% in terms of the Brier Score, and 10.8% in terms of R^2 . Put differently, the machine learning model is better able to predict the racial and ethnic identities of borrowers using observable characteristics.⁴⁴ Whether this ability contributes to triangulation will depend on whether there is considerable variation in true default propensities across race and ethnic groups that is unrelated (or at least unrelated up to the detection capabilities of the more sophisticated nonlinear model) to observable characteristics. We consider this issue in more detail when we compute bounds on triangulation and flexibility.

Next, we document how estimated PDs from these models vary across race groups in U.S. mortgage data, to better understand the differences in distributional consequences between the traditional and sophisticated models.

⁴⁴ Interestingly, if we rerun this exercise including the interest rate R as an additional predictor, we are able to predict race even more precisely using the RF model, with an R^2 improvement of 30% relative to NL.

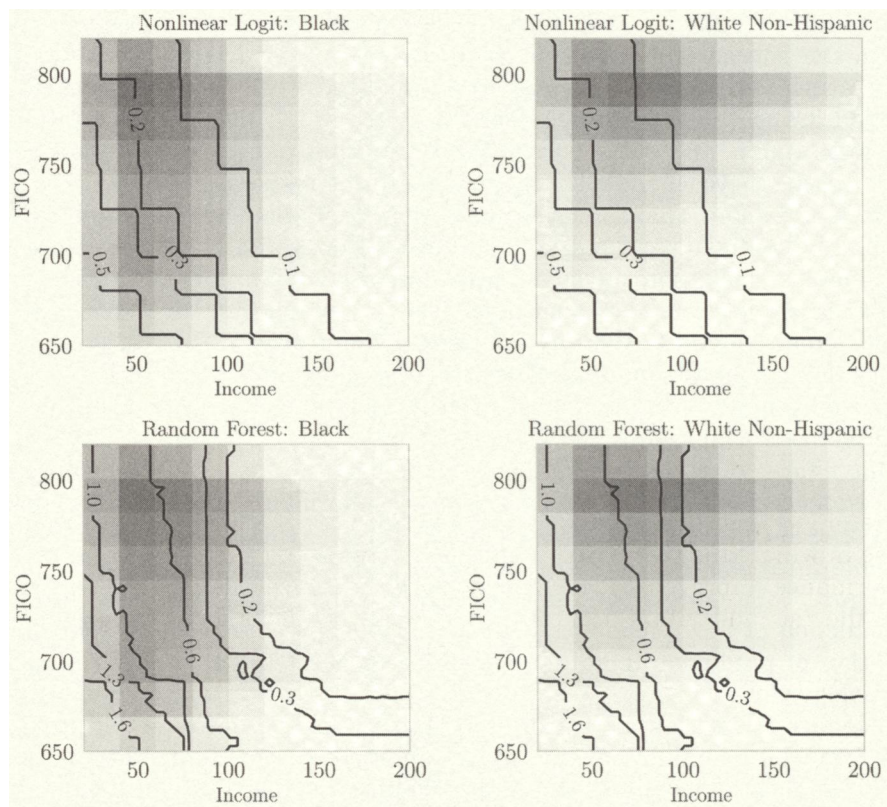


Figure 4. Example of predicted default probabilities across models. This figure shows level sets of default probabilities (in %) predicted from different statistical models for different values of borrower income and FICO (holding other characteristics fixed as explained in text). NL predictions are shown in the top row; RF predictions are in the bottom row. Heatmaps show the distribution of borrowers within certain race/ethnicity groups: Black in left column and White non-Hispanic in right column. (Color figure can be viewed at wileyonlinelibrary.com)

D. Differences in Predicted Default Propensities

We begin our analysis of distributional consequences with Figure 4. This figure, which is based on a subset of the data, illustrates the estimated models and extends our motivating theoretical example in Figure 1 to two dimensions. Each panel of the figure plots level sets (or isoquants) of predicted default propensities as a function of borrower income on the horizontal axis and FICO score on the vertical axis. We fix other borrower characteristics in the subset of the data, and then obtain predicted default probabilities from our estimated models by combining this fixed constellation with every income-FICO combination on a two-dimensional grid.⁴⁵ The level sets of predicted default

⁴⁵ Specifically, we vary income and FICO for loans originated in California in 2011 with a loan amount of 300,000 U.S.\$, LTV of 80, term of 30 years, and purpose of buying a home. We focus on loans issued to owner-occupants with full documentation and securitized through Fannie Mae.

probabilities for the NL model are in the top two panels and those for the RF are in the bottom two panels. These level sets are overlaid with a heatmap illustrating the empirical density of income and FICO levels among minority (Black and White Hispanic) borrowers in the left panels and White non-Hispanic, Asian, and other borrowers in the right panels, with darker colors representing more common characteristics in the respective group.

The figure shows that the level sets of the RF predicted default probabilities are highly nonlinear and markedly different from those of the NL. Moreover, they increase more sharply as we move into the low-income, low-FICO region in the bottom left corner of each chart. The heatmaps show that these regions are also relatively more densely inhabited by minority borrowers. This graphical example suggests (similar to our theoretical example) that at least for the restricted sample that we consider in these plots, the effects of new technology differ across racial groups.

This figure, while intriguing, needs to be backed up with more comprehensive evidence from the full data set. In particular, many “leaves” of our underlying tree models (e.g., leaves associated with LTV ratios below 70) and many possible interactions (e.g., between FICO and LTV) never come into play in the figure.⁴⁶ To more rigorously assess who wins versus who loses from the introduction of new technology, we next study the change in the entire distribution of predicted default propensities in our test sample as estimation technology varies.

In Figure 5, we look at the differences between the entire set of predicted PDs from the machine learning model and those from the traditional Logit model. Panel A of the figure shows the cumulative distribution function (cdf) of the difference in the estimated default probability (in percentage points) between the RF and NL. Borrowers for whom this difference is negative (i.e., to the left of the vertical line) are “winners” from the change to the new technology, in the sense of having a lower estimated default probability, whereas those with a positive difference (i.e., to the right of the vertical line) are “losers.” For each level of difference in PDs across the two models on the *x*-axis, the *y*-axis shows the cumulative share of borrowers at or below that level; each line in the plot shows this cdf for a different race/ethnic group. Panel B plots the log difference in PDs to highlight the *proportional* benefit for each group.⁴⁷

Both panels show that there is a reduction in default risk under the RF model for the median borrower (indicated by the dashed horizontal lines) in the population as a whole. Indeed, the plot shows that for *all* groups, the share of borrowers for whom the estimated probability of default drops under the new technology is above 50%.

⁴⁶ In addition, for the particular constellation of characteristics, we consider in Figure 4, the RF predicts uniformly higher default probabilities than Logit. This difference seen in the given subset of the data (used for illustrative purposes) is not representative of the overall population, however, where the sample average predicted probability of default is 0.74% for both RF and Logit predictions, which also coincides with the true average probability of default in our data.

⁴⁷ For ease of visual representation, we truncate the *x*-axes on these plots, as there exists a small fraction of cases for which the estimated differences in the default probabilities are substantial.

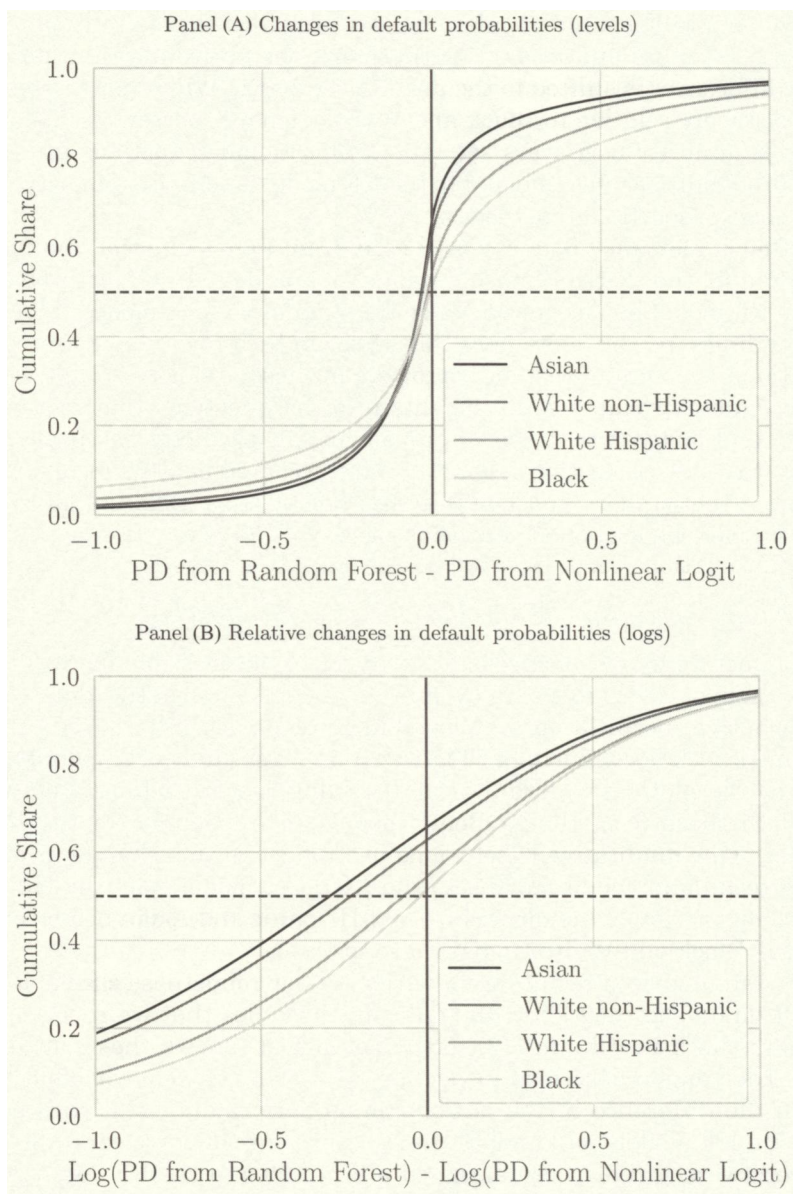


Figure 5. Comparison of predicted default probabilities across models, by race groups. (Color figure can be viewed at wileyonlinelibrary.com)

However, this graph also highlights the important *differences* in the outcomes arising from the new technology between race groups. Panel B clearly shows that the winners from the new technology are disproportionately White non-Hispanic and Asian—the share of borrowers in these groups that benefit from the new technology is roughly 10 percentage points higher than for the

Black and White Hispanic populations, within which there are roughly equal fractions of winners and losers. Furthermore, the entire distribution of relative PD differences is shifted to the north-west for the White non-Hispanic and Asian borrowers relative to Black and White Hispanic borrowers. This means that fewer minority borrowers see large proportional reductions in predicted default probabilities when moving to the RF model, and more minority borrowers see large proportional increases.

A further important takeaway especially from Panel A is that for these minority groups, the distribution of changes in predicted default probabilities from the RF model has larger variance than under the NL model.⁴⁸ We return to this finding later, but note here that this can be interpreted as introducing an additional element of risk for borrowers in these minority groups.

These figures provide useful insights into the questions that motivate our analysis, and suggest that the improvements in predictive accuracy engendered by the new prediction technology are accompanied by an unequal distribution of the winners and losers across race groups. Before analyzing this finding further, we first check its robustness.

E. Alternative Specifications and Robustness of the Models

We conduct a number of checks to verify the robustness of our results to alternative sets of covariates employed in the models, alternative data samples, and alternative specifications employed in our comparisons.

Our first check is conducted in Figure IA.3 in the Internet Appendix, where we repeat the plots in Figure 5 after replacing the predictions from the RF model with those from the XGBoost model (also explained in the Internet Appendix). Our qualitative conclusions are robust to this change in machine learning technique: when moving to the XGBoost model from the NL model, there are more winners among White non-Hispanic and Asian borrowers than among the Black and White Hispanic groups.

Table V summarizes results from various other robustness checks in a consistent format. The rows show the outcome variables that we report for each robustness check in columns (2) to (8); column (1) shows these outcomes in the baseline version described in Section III.C. The outcome variables are, in Panel A, the difference in the out-of-sample performance statistics between the RF and NL models; in Panel B, the percent share of borrowers who appear more creditworthy (i.e., those assigned a lower PD) in the RF versus the NL model, to summarize the differences detected between groups in Figure 5; and in Panel C, the standard deviations of the changes in the PDs between the two models, to summarize the variance in PDs faced by a borrower in each group when moving from the traditional to the more sophisticated technology.

Column (2) adds interest rates to the set of covariates. We find no substantial changes to the baseline outcomes arising from this addition. Column (3)

⁴⁸ The distributions are also right-skewed, that is, the RF model predicts far higher PDs than the Logit model for some borrowers in all groups.

Table V
Alternative Models and Subsamples of Data

This table shows improvements in performance metrics when going from Nonlinear Logit to Random Forest (Panel A), the percentage of borrowers (by race group) who are assigned a lower PD by the Random Forest than by the Nonlinear Logit (Panel B), and the standard deviation (by race group) of changes in PD when going from Random Forest to Nonlinear Logit (Panel C), for different model versions or estimation samples, as described in the text. The last two rows show for each column the number of observations used for the estimation of the model and the number of observations used for out-of-sample evaluation (testing).

	Change Covariates			Alternative Samples			Change Spec.	
	Baseline (1)	Int. Rates (2)	No Fico (3)	No Unknowns (4)	2009 to 2011 (5)	White (6)		GSE+Full Doc (7)
Panel A: Percentage Difference in Performance Statistics between Random Forest and Nonlinear Logit								
ROC	0.77	0.76	1.64	0.73	0.62	0.60	0.58	0.79
Precision	5.06	4.83	11.57	4.33	1.34	3.36	1.56	2.70
Brier Score	-0.40	-0.45	-0.30	-0.41	-0.40	-0.35	-0.30	-0.66
R ²	14.30	15.64	21.28	14.78	12.06	13.71	11.74	25.66
Panel B: Percent Share of Borrowers Who Look More Creditworthy in Random Forest versus Nonlinear Logit								
White Non-Hisp.	12.68	13.07	5.59	12.68	12.69	11.60	9.33	7.10
Asian	15.62	16.14	5.09	15.76	15.44	-	12.00	8.57
Black	1.29	1.04	-0.49	1.48	1.58	-	1.50	-1.44
White Hispanic	4.40	3.77	-0.00	4.54	6.47	-	2.91	-0.30
Panel C: Standard Deviation of Δ Probability of Default between Random Forest versus Nonlinear Logit								
White Non-Hisp.	0.0071	0.0072	0.0057	0.0071	0.0082	0.0073	0.0061	0.0067
Asian	0.0061	0.0061	0.0052	0.0062	0.0074	-	0.0057	0.0057
Black	0.0136	0.0139	0.0084	0.0137	0.0161	-	0.0120	0.0122
White Hispanic	0.0102	0.0103	0.0069	0.0104	0.0128	-	0.0089	0.0092
# Obs. used for estimation	4,591,292	4,591,292	4,591,292	4,108,819	2,757,163	3,495,156	2,522,670	4,591,292
# Obs. used for testing	2,810,996	2,810,996	2,810,996	2,515,826	1,687,444	2,140,187	1,545,697	2,810,996

reports results when we remove FICO from the baseline covariates. The performance differences between the machine learning model and the Logit model here are more substantial, suggesting that the FICO score already contains some of the information gain arising from nonlinearly combining the other covariates. However, while attenuated, it is still the case that the majority White group gains under the machine learning model in terms of lower average PDs, and the standard deviation of PDs continues to be higher for the minority groups under the machine learning model.

The next four columns consider changes to the sample and their effects on our inferences. Column (4) removes the observations with “Unknown” race (10.5% of our sample); column (5) reruns our analysis using only the 2009 to 2011 period during which the market share of FinTech lenders (taken from Buchak et al. (2018)) was very low (2% to 3% of originated loans); column (6) reruns our analysis only for the majority White group; and column (7) restricts the sample to GSE-backed loans with full documentation (a subsample that we return to in the next section). Columns (4), (5), and (7) show that our inferences remain similar despite changes to the sample, and column (6) shows that the performance differences are also of a similar magnitude across the machine learning and traditional models when we focus only on the majority group.

Finally, in column (8) of Table V, we use a more flexible version of the NL: we fully interact the FICO and LTV bin indicators with one another, and we further interact the resulting terms with separate dummies indicated whether the loan is used for a home purchase, whether the loan is underwritten with full documentation, and whether the loan has a duration of 15 years or less. If this added flexibility improves the predictive performance of the Logit, we would expect the performance metric differences in Panel A to decrease relative to column (1). We see, however, that this is only the case for precision, but not the other three metrics. These findings indicate that according to the ROC-AUC, R^2 , and Brier Score metrics, the more complex version of the NL actually performs worse out of sample than our baseline version. This is perhaps unsurprising, given the fact that the more complex Logit model is likely to suffer more from overfitting. In contrast, the RF model naturally controls the variance of the out-of-sample predictions.

Having confirmed that the differences between groups across the different models are robust, we seek a better understanding of the underlying sources of these unequal effects. We turn to this question next.

F. Flexibility and Triangulation in the Data

In Section I, we proposed that better statistical technology can have differential effects on different groups in the population due to increased *flexibility* to learn nonlinear combinations of characteristics that directly predict default and/or increased ability to use these nonlinear combinations to *triangulate* hidden variables such as race. In this section, we construct simple empirical measures in an attempt to gauge the relative importance of these two mechanisms.

Table VI
Decomposition of Performance Improvement

This table shows the decomposition of improvement for different model performance metrics, as described in the text.

Panel A: Race Controls First			
	Race	Race Int.	Technology
ROC-AUC	6.28	2.04	91.69
Precision	9.05	22.43	68.52
R^2 /Brier Score	3.37	4.39	92.24
Panel B: New Technology First			
	Technology		Race
ROC-AUC	89.77		10.23
Precision	94.14		5.86
R^2 /Brier Score	92.95		7.05

Consider the performance of the NL and RF models reported in the second and third rows of Table III. If we take the NL without race as a baseline scenario, the greatest increase in predictive power relative to this model will clearly be achieved by both employing a better technology and allowing for the use of race, that is, by using the RF with race. For example, for the R^2 reported in the final two columns of the table, this performance improvement is about 15.6% ($\frac{0.0318-0.0275}{0.0275}$).

Our goal is to bound the extent of this total improvement arising from flexibility relative to that from triangulation. This is a nontrivial task because, in some cases, the decomposition we propose may not generate a meaningful distinction between triangulation and flexibility. For example, suppose that the true default probability is $y = f(x) + \varepsilon$ for a nonlinear function $f(x)$ of observable characteristics, but $f(x)$ is perfectly correlated with group indicators g . In this case, there is no meaningful distinction between a technology that flexibly estimates $f(x)$ and one that triangulates g to predict default. Given this identification problem, we do not pursue exact measures of triangulation and flexibility. Rather, we seek to provide bounds from our empirical results on the relative importance of flexibility and triangulation. In the Internet Appendix, we further validate our measures by analyzing a simple structural model in which triangulation and flexibility have natural and precise definitions. In the setting of that model, we show that the empirical measures that we propose here are indeed bounds on “true” underlying flexibility and triangulation. Our discussion below interprets these measures accordingly.

As in any partial regression exercise, there are two possible ways to arrive at increases in explanatory power, depending on the order in which variables enter the model. We report results of this exercise based on both of these approaches in Table VI. In Panel A, we add in race dummies first, fixing the NL as the baseline statistical technology. The left column reports the percentage of

the overall increase in the performance that can be achieved by simply adding race dummies to the set of variables available to the NL, without changing technologies. For example, around 3.4% of the total performance improvement (15.6%) in terms of R^2 or Brier Score arises from the inclusion of race dummies as covariates in the NL model.⁴⁹ In the second column of the table, we allow for the addition of race dummies to the NL in a more flexible manner, interacting them with all other borrower/loan characteristics. This improves the performance metrics even further, with an incremental gain of 4.4% in R^2 . (The gain is even higher for the Precision measure). Finally, the rightmost column of Panel A shows 100% minus the sum of the fractions in the previous two columns, which gives the fraction of the total performance improvement from moving to the RF over and above the improvement achieved by simply adding race in naïve and more flexible ways. For example, moving from the NL model with race interactions to the RF model (also with race) delivers roughly 92% of the 15.6% total improvement in R^2 .

In Panel B, we add new technology first, fixing x (i.e., without including race) as the vector of explanatory variables. The leftmost column in this panel shows the fraction of the overall improvement that is achieved by changing technology (moving from NL to the RF, without including race in either model), while the right column shows its complement (i.e., 100% minus the “Technology” improvement), attributable to improvements in the performance from adding race conditional on already having a flexible model (i.e., moving from RF without race to RF with race). For two of three performance measures, a larger fraction of the improvement is now attributed to race conditional on the new technology than in the NL model. This is not too surprising, as additional interactions between race and other observables can be used by the RF. This result suggests that machine learning models are able to capitalize on interaction effects between race and other characteristics.

The table yields several interesting observations. First, if $f(x)$ were perfectly correlated with g —the “unidentified” case referred to earlier—the left columns in Panels A and B would both show 100%, meaning that it would be impossible to tell whether the predictive improvements that we find arise from flexibility or triangulation. This is clearly not the case in our empirical estimates, which consistently imply that a larger share of the increase in accuracy stems from the more flexible technology than from the inclusion of race dummies. This suggests that when predicting mortgage default, triangulation alone does not explain the major share of the performance improvements from machine learning, although triangulation does emerge as a nontrivial factor according to all metrics and as a substantial factor according to one of the three performance metrics (i.e., Precision). Indeed, the numbers in the first and second columns of Panel A suggest that knowing race (which is the best that triangulation without additional flexibility could achieve) would yield up to 31.5% of the total

⁴⁹ Note that the decomposition for R^2 and the Brier Score are identical due to their arithmetic relationship. Since $R^2 = 1 - (\text{Brier} / (\text{Pr}(\text{Default}) \times (1 - \text{Pr}(\text{Default}))))$, any percentage comparison of R_i^2 and R_j^2 where $\text{Pr}(\text{Default})$ is fixed is identical to comparing the Brier Scores.

performance improvement, although the averages across performance metrics suggest smaller increases.

Note that Panel B is not as informative about the share of the overall improvement that can be attributed to flexibility, since the improvements generated by the move from the Logit to the RF model could stem from either flexibility or triangulation. The high share of the performance improvement arising from this move simply provides an upper bound on what can be achieved by having more flexibility in the model. When predicting mortgage default in our sample, we find that this upper bound is large. That said, we note that in other applications or indeed in other samples, it may be tighter and suggest larger effects of triangulation.

The bounds reported in this section have implications for policy. Current regulations forbid the use of race as a predictive variable. If we found that a large proportion of predictive improvements could be attributed to triangulation, this would suggest that new statistical technology could be viewed as regulatory arbitrage, as it effectively reintroduces race into the model. In this case, an appropriate policy response might be to discourage the use of such technology in household credit markets.

By contrast, our results suggest that the main effects in our empirical setting arise from the fact that existing regulations cannot protect minorities from the additional flexibility conferred by the new technology. This flexibility allows new technology to map distributions of characteristics such as income and FICO scores (themselves affected by unobservable, potentially historic determinants correlated with group status) to default probabilities. The policy implications of these results are different from those in the case of triangulation. The fact that unequal effects in this exercise appear to be driven mainly by flexibility does not make them less unequal. While a full welfare analysis is beyond the scope of this paper, it may remain desirable for a regulator to counteract the unequal effects of new technology if she has a sufficiently high preference for equality of access to credit. Given the importance of flexibility, it would then become necessary to go beyond current regulations to consider the broader mapping between borrower characteristics and credit scores.

IV. Illustration of Equilibrium Effects

Our discussion so far focuses on the impact of machine learning technology on predicted PDs across different groups of borrowers. In credit markets, these effects can translate into changes in lenders' decisions both on the extensive margin (i.e., which borrowers to accept for credit versus which to reject) and the intensive margin (i.e., how to price loans to borrowers who are accepted). To arrive at a tractable "back-of-the-envelope" computation of how these effects on default probabilities might map to these final outcomes, we conduct two exercises. Our main approach, described in this section, consists of a

simple calibrated one-period equilibrium model.⁵⁰ As a reduced-form alternative, we empirically estimate a mapping from default probabilities to interest rates under the assumption that observed interest rates in our sample were generated by the Logit technology. We then study how the distribution of rates across groups would change if lenders moved to RF technology but held the mapping constant. The results from this exercise, described in detail in the Internet Appendix, are qualitatively consistent with what follows.

A. A Simple Model of Mortgage Market Equilibrium

Our model is not intended to be used for full structural estimation, but rather to illustrate the potential effects of new statistical technology. To facilitate this exercise, we make three strong assumptions.

First, we assume that the market for mortgages is competitive. Lenders set interest rates R contingent on borrowers' observable characteristics x . Lenders are risk-neutral and discount future repayments at rate ρ . All nonprice characteristics of mortgage contracts, such as the loan amount and LTV ratio, are predetermined for each borrower.⁵¹

Second, we assume that lenders have access to a statistical technology that provides an unbiased estimate $\hat{P}(x, R)$ of the structural probability of default when a borrower has characteristics x and accepts a mortgage with interest rate R . The dependence of $\hat{P}$ on R captures, in reduced form, the fact that a higher required monthly payment increases the likelihood that a borrower finds it beneficial to default on their loan (e.g., after a negative income shock); we discuss potential problems with estimating the relationship between default and R below. Lenders in the model are fully rational, but constrained in their mapping of characteristics and rates to default probabilities by the available statistical technology. As technology varies, the mapping that lenders use also changes—this allows us to plug the estimated functions $\hat{P}(x, R)$ from various technologies into equilibrium conditions to arrive at our back-of-the-envelope estimates.

Third, as discussed earlier, we assume no possibility of prepayment. Each mortgage is either repaid in full at the end of its term, for a repayment of $1 + R$ per dollar of the loan, or ends in default, for a repayment of $1 + R - LGD(x, R)$. Here, $LGD(x, R)$ stands for the loss given default for a borrower with characteristics x , which we assume to be deterministic.

⁵⁰ The Internet Appendix contains a more rigorous analysis of this model, including a basic discussion of how the analysis might change if we were to instead consider the steady state of a dynamic model of credit provision rather than a one-period model.

⁵¹ In reality, these parameters are often dictated, or at least confined to a narrow range, by local property prices and borrower liquidity constraints. In the Internet Appendix, we discuss how endogenous contracting terms would affect our results. Our focus on a single price for a given contract is motivated mainly by tractability, consistent with a large applied literature on insurance contracts (e.g., Einav, Finkelstein, and Levin (2010)).

Under these assumptions, each lender estimates net present value (NPV) of offering a mortgage with rate R , per dollar of the loan amount, as:

$$N(x, R) = \frac{1}{1 + \rho} [1 + R - \hat{P}(x, R) \cdot LGD(x, R)] - 1. \quad (4)$$

In a Bertrand equilibrium, borrowers with characteristics x are offered the lowest interest rate $R(x)$ that satisfies the break-even condition $N(x, R(x)) = 0$. Because default probabilities depend on interest rates, it is possible that no break-even rate exists. Intuitively, this has the flavor of market unravelling. If default risk $P(x, R)$ is high and if, in addition, it increases in R , then lenders cannot break even at low interest rates, but they also cannot break even by charging higher rates, since doing so would exacerbate default risk. In that case, we say that borrowers with characteristics x are rejected for a loan.

B. Empirical Implementation

To explore the equilibrium implications of machine learning in the data, we solve for equilibrium rates $R(x)$ (or rejection, when no breakeven R exists) implied by different statistical technologies for borrowers in our data set. Specifically, we plug predicted default probabilities $\hat{P}(x, R)$ from both traditional technology and machine learning models into equation (4). We further calibrate the lender's discount rate ρ and the loss given default $LGD(x, R)$ directly based on empirical data from the mortgage market. Finally, we solve the equation $N(x, R(x)) = 0$ to find the equilibrium (break-even) interest rate. The Internet Appendix describes the computational approach and our calibration in detail.

This approach requires that we rerun the predictive exercise above using models that also include the interest rate R as a predictive variable. While our empirical results do not differ greatly between models that do and do not contain observed interest rates (as shown in Table V), adding R is conceptually questionable because interest rates in our data are not randomly assigned. To correctly estimate the sensitivity of default probabilities to interest rates (even in a machine learning model), one needs to assume *no selection on unobservables*, that is, interest rates need to be assigned independently of default outcomes *conditional on controlling for observable characteristics* x . We state the conditions required for identification more formally in the Internet Appendix, where we also discuss the related issue of adverse or advantageous selection by borrowers.

In reality, a key threat to identification is that interest rates could partly reflect lenders' "soft" information about borrowers—which is unobservable from our perspective.⁵² Accordingly, we take a two-pronged approach to try to address this issue and to obtain results that are robust even if the assumption

⁵² Strictly speaking, the fact that there is variance in R (conditional on x) in the data is also at odds with the Bertrand competition assumption in the model. In practice, R contains "noise" that is indeed plausibly orthogonal to x , even if the market is competitive—for example, interest rates often vary even intraday based on developments in global bond markets that are exogenous to any

above is not fully satisfied in the data. First, when estimating default probabilities that feed into equilibrium computations, we include only GSE-insured mortgages (i.e., those securitized through Fannie Mae or Freddie Mac) that are marked as having been originated with full documentation of borrower income and assets—Table V, column (7), shows how our results on PDs look in this sample. In this segment, soft information is less likely to be important because lenders focus mainly on whether a borrower fulfils the underwriting criteria set by the GSEs.⁵³

Second, we rely on and extend existing work by Fuster and Willen (2017) to adjust our estimates of $\hat{P}(x, R)$ for any residual bias. We do this by estimating the bias as the difference between causal and noncausal estimated effects of interest rate changes on mortgage default.⁵⁴

We finally note that in our data, we cannot observe borrowers not granted mortgages, and thus, we restrict our counterfactual statements to populations with distributions of characteristics identical to the one we observe. Under the assumption that borrowers denied a mortgage are high credit risks, we therefore potentially understate (overstate) the population averages of extensive margin credit expansions (contractions) when evaluating equilibrium under a counterfactual technology. On the other hand, an advantage of focusing on GSE loans, in addition to soft information playing little role in underwriting and pricing, is that GSE policies likely led to more loan originations than would have obtained in a purely private market. This helps reduce concerns about extensive margin biases.

C. Equilibrium Results

Table VII summarizes our computation of lending and pricing decisions from the model. The first two columns show the average acceptance rate for the NL model and the RF model. The third and fourth columns show the average spread (SATO) charged to borrowers conditional on acceptance, and the final two columns show the dispersion of spreads conditional on acceptance. The first five rows of the table present these statistics for each of the racial groups in the data, and the sixth row presents results averaged across the entire population. The final row shows the standard deviation of average acceptance rates and spreads across racial groups (this is the cross-sectional spread of the group means relative to the population mean, where each group is weighted by its share in the sample).

given borrower taking out a loan that day. Moreover, it is clearly the case that a more realistic model might depart from Bertrand competition to incorporate search and other frictions.

⁵³ In influential work, Keys et al. (2010) argue that there are discontinuities in lender screening at FICO cutoffs that determine the ease of securitization, but only for low-documentation loans (where soft information is likely more important)—not for full-documentation loans that we consider.

⁵⁴ The adjustment is described in detail in the Internet Appendix, which also describes other relevant details of our equilibrium calculations.

Table VII

Equilibrium Outcomes

This table shows predicted acceptance rates, average interest rates (expressed as “SATO” or spread at origination), and the standard deviation in SATO for different groups under the assumption that the statistical model used for default prediction is the Nonlinear Logit (NL) or the Random Forest (RF).

	Accept (%)		Mean SATO (%)		SD SATO (%)	
	(1) NL	(2) RF	(3) NL	(4) RF	(5) NL	(6) RF
Asian	92.4	93.3	−0.108	−0.123	0.274	0.322
White Non-Hispanic	90.3	91.1	−0.083	−0.090	0.296	0.356
White Hispanic	85.6	86.4	−0.031	−0.008	0.333	0.414
Black	77.7	79.3	0.022	0.060	0.365	0.461
Other	88.9	89.5	−0.083	−0.088	0.296	0.360
Population	89.8	90.7	−0.081	−0.086	0.298	0.360
Cross-group SD	2.165	2.098	0.020	0.029		

We find that the proportion of borrowers in the population that are accepted for a mortgage increases by about 0.9 percentage points when lenders use RF instead of Logit.⁵⁵ This increase benefits all racial groups and is particularly pronounced for Black borrowers. Perhaps, intuitively, the superior technology is better at screening, and is therefore more inclusive on average, and in a manner that cuts across race groups. Consistent with this intuition, the cross-group standard deviation of acceptance rates decreases for this model.

The average interest rate spread when moving to the more sophisticated model falls slightly (by 2 basis points (bps)) for Asian borrowers and increases (by 4 bps) for Black borrowers. The cross-group standard deviation of spreads increases by almost 50% relative to its baseline value of 2 bps under the Logit model. That is, average pricing effects implied by our simple model are also unequal across race groups, consistent with our results on predicted default probabilities in Section III. While a 6 bps differential change in the interest rate has only a modest effect on the resulting interest payment—about \$120 over the first year of a typical new mortgage⁵⁶—the total effect across groups is still not negligible, given the size of the U.S. mortgage market.⁵⁷

⁵⁵ In the Internet Appendix, we further show that the aggregate increase in acceptance rates masks some heterogeneity: about 1.7% of the population is newly excluded when moving to the machine learning model, while about 2.6% is newly included.

⁵⁶ Assumption: 30-year fixed-rate mortgage over US\$200,000 with baseline interest rate of 4.5%.

⁵⁷ For instance, over our sample period from 2009 to 2013, 1.8 million mortgages were originated by Black borrowers according to HMDA data. Taking the flow of originations over these five years as a rough approximation to the stock of outstanding loans, a US\$100 difference in annual interest payments would correspond to a total cost to Black borrowers of US\$180 million per year. Using a slightly different calculation, Bartlett et al. (2021) estimate that an increase in the interest rate paid by Latinx and African-American borrowers of 4.6 bps—corresponding to Bartlett et al.

We see larger effects of the more sophisticated technology on the dispersion of interest rates both in the overall population and within each group (columns (5) and (6)). These results are reminiscent of our Lemma 1, in which the new technology generates predictions that are a mean-preserving spread of the older technology. Overall, we compute that the dispersion in rates increases by about 21% ($= \frac{0.360-0.298}{0.298}$).

The cross-group variation in the within-group *dispersion* of rates is also very different across the models. The breakdown by racial groups reveals that the increase in dispersion is much more pronounced for minority borrowers: the standard deviation of interest rates increases by 5 and 6 bps for Asian and White non-Hispanic borrowers, respectively, but by 8 and 10 bps for White Hispanic and Black borrowers. The *proportional* increases in within-group dispersion are also substantially larger for these minority groups. These patterns suggest that the RF model screens within minority groups more extensively than the Logit model, leading to changes in intensive margin lending decisions associated with the new technology. It also suggests an important form of risk confronting White Hispanic and Black borrowers, namely, that their rates are drawn from a distribution with higher variance under the new technology. This introduces an additional penalty for risk-averse borrowers.

Overall, we obtain an interesting picture. As we see above, the RF model is a more accurate predictor of defaults. Moreover, it generates higher acceptance rates on average. However, it penalizes some minority race groups significantly more than the previous technology, leading them to receive higher and more disperse interest rates. To take the next step relative to the simple exercise we conduct here, and to arrive at a more realistic set of magnitudes, one would need (i) purely random variation (rather than the bias corrections that we employ) in interest rates to correctly estimate $\hat{P}(x, R)$ and calibrate equilibrium outcomes given a baseline data set; (ii) a dynamic model of borrower behavior in which we can simulate a new cohort of data from first-round outcomes to recalibrate equilibrium, and so forth, ideally until convergence; and (iii) observations on the behavior of borrowers not granted mortgages to understand the full counterfactual effects under the new technology. We discuss several of these issues in the Internet Appendix in greater detail.

V. Conclusion

In this paper, we analyze the distributional consequences of changes in statistical technology used to evaluate creditworthiness. Our analysis is motivated by the rapid adoption of machine learning technology in this and other financial market settings. We find that these changes in technology can increase disparity in credit market outcomes across different groups of borrowers in the economy—based, for example, on race. We present simple theoretical frameworks to provide insights on the underlying forces that can drive

(2021)'s estimated weighted average discrimination across purchase and refinance loans—costs these borrowers US\$765 million per year.

such outcomes. Then, we provide empirical evidence, suggesting that these issues arise in U.S. mortgage data, focusing on the distribution of mortgages and rates across race-based groups.

The main insight of our paper is that a more sophisticated statistical technology leads to more dispersed predictions as it better fits the predicted outcome variable (in our setting, the probability of mortgage default). It immediately follows that such dispersion will generate both “winners” and “losers” relative to their position in equilibrium under the preexisting technology.

Using a large data set from the U.S. mortgage market, and evaluating a change from a traditional Logit technology to machine learning technologies, we find that Black and White Hispanic borrowers are predicted to lose, relative to White and Asian borrowers. This is true both in terms of the distribution of predicted default propensities, and based on a simple equilibrium model that we construct to ascertain magnitudes, in terms of counterfactual interest rates—not only on average, but also in terms of larger within-group dispersion.

We propose two possible mechanisms through which such distributional changes could come about. The first arises from the increased flexibility of the new technology to better capture the structural relationship between observable characteristics and default outcomes. The second arises from the improved ability of new technology to more effectively triangulate borrowers’ (hidden) identities. With an improved ability to triangulate, and if these groups have higher default probabilities even controlling for observables, the new technology might penalize certain groups of borrowers over and above what the structural relationship between other observables and default outcomes might predict. We suggest a simple way to bound the relative importance of these two mechanisms, and find in our empirical application that while under some metrics both flexibility and triangulation play an important role, flexibility is the primary source of unequal effects between groups.

Clearly, increases in predictive accuracy can (and in our setting, do) arise from new technologies’ improved use of information. However, our work shows that at least one reason to more carefully study the impact of introducing such technologies is that the winners and losers from the widespread adoption of these new technologies can be unequally distributed across societally important categories such as race, age, or gender. Our work provides early evidence on these impacts in the domain of credit markets. We believe that much more work needs to be done to understand the impacts of these technologies on the distribution of outcomes in a wide variety of financial and goods markets.

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Appendix: Proof of Lemma 1

We denote by $\mathcal{L}^2$ the space of random variables z such that $E[z^2] < \infty$, where the inner product $\langle x, y \rangle = E[xy]$. Let $P(x) = E[y|x]$ and assume that $P(x) \in \mathcal{L}^2$.

Let $\hat{P}_j$ denote the projection of P onto a closed subspace $\mathcal{M}_j \subset \mathcal{L}^2$. It is well known (e.g., chapter 2 of Brockwell and Davis (2006)) that the projection $\hat{P}_j(x)$ minimizes the MSE $E[(P(x) - \hat{P}_j(x))^2]$. Hence, the projection coincides with the ideal predictor defined in (1):

$$\hat{P}_j(x) = \hat{P}(x|\mathcal{M}_j).$$

We need to show that $\hat{P}_2$ is a mean-preserving spread of $\hat{P}_1$, that is, that $\hat{P}_2 = \hat{P}_1 + u$, where $E[u] = 0$ and $\text{Cov}(u, \hat{P}_1) = 0$. By the projection theorem, we have

$$E(m, P - \hat{P}_j) = 0$$

for any $m \in \mathcal{M}_j$. Letting $m \equiv 1$, we obtain $E[\hat{P}_j] = E[P]$, which directly gives us $E[u] = 0$. Moreover, note that

$$\text{Cov}(u, \hat{P}_1) = \text{Cov}(\hat{P}_2 - P, \hat{P}_1) + \text{Cov}(P - \hat{P}_1, \hat{P}_1).$$

The first term is zero by an application of the projection theorem to $\hat{P}_2$, noting that $\hat{P}_1 \in \mathcal{M}_1 \subset \mathcal{M}_2$. The second term is zero by a direct application of the projection theorem to $\hat{P}_1$.

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Supporting Information

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Appendix S1: Internet Appendix.
Replication Code.